P. P. Dept Meeting — April 13, 1959

Multiprogramming — Codd.

1. Historical Development

Steps toward simultaneous processing — a trend in computer design.

(1) (701, 702, 704) one job at Time
— idle Time

Example: punched card installation: at one time

| | | |
|---|---|---|
| sorter | collator | } all busy |
| Tabulator | Interpeter | |
| | Reproducer | } not busy |

701 Installation: — only one tape running

(2) 705: limited simultaneous — file maintenance.
one read or write overlapped with calc.
— End of r-w op not easily doable.
— expensive

(3) 705 - 709 with DSU. — more general & flexible
— up to 6 simult. ops. — read may overlap read, etc.
— still have trouble with end of op being RPQ'd.
(usually 705's have 2 TRC)

(4) 7070, 7090, 7050
— interrupt on I/O. same as above otherwise

(5) STRETCH

5 possible upgrade from I/O.

(6) BULL GAMMA 60.

Scientific processing extended to internal units.

- Decimal arith. Unit
- Logical (binary unit)
- Comparator
- input units

less lack of ~~all~~ interrupt & HS memory at present

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Multi-facility machine :

uniprogramming — execution of programs in sequence.

Interleaving : — switching of facility from one prog to another.

Overlapping : — simultaneous execution of progs. on diffent facilities.

Multiprog : — either or both interleaving or overlapping.

Reasons for Multiprog :

1. greater efficiency on batch processing.
2. Console debugging
3. Remote inputs from indep. timed
4. application of computers - close coop. between man & machine

Example of Batch processing:

Prog A: 16K wds data, 2K instr, 2M wds backup

CPU: 720 ms
Disk write
Disk read

(Mesh problem)

Prog B: card-tape (editing – conversion – grouping)
1K wds .5K instr

Read Cds 12000 ms
CPU 20
Write Tape 120 ms

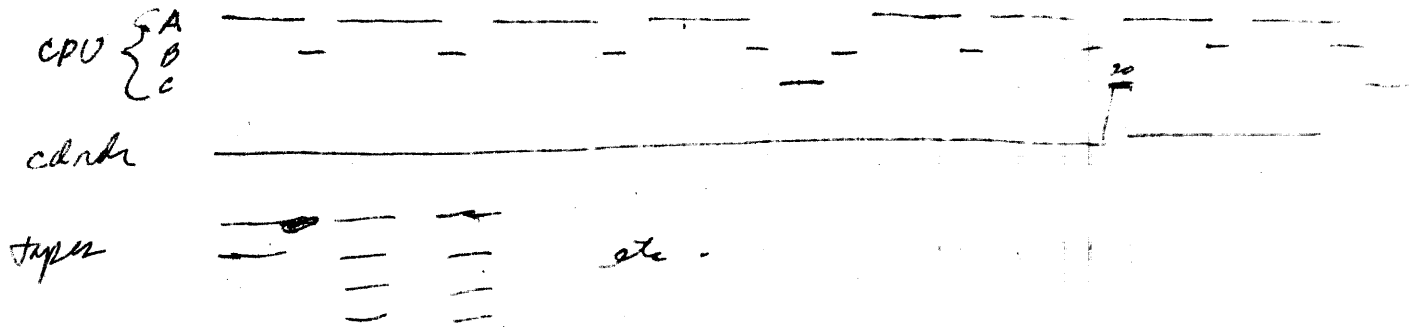
Prog C: file maint 4K data 3½K instr 4 Tapes

Read Tape 1 150 150
" Tape 2
CPU 50 50
Write Tape 3
Write Tape 4

Total: Mem 27K wds, Disk 2M wds, Cdrdr, 5 Tapes

together:

Rule: displace prog if another prog uses CPU less freq or for shorter length of time.



If each by itself ran for 2 hours.
Total separately would be 6 hrs.

Prog A : is delayed 30%
B : is not delayed - except negligible extension.
C : no delay at all.

So total = 2 hr 35 min favorable example.

Example: Two programs of same characteristics. A & A'

| | A | A' | A | A' | A | A' |
|---|---|---|---|---|---|---|
| CPU | | | | | | |
| Disk retrieve write | — | — | — | — | | |
| | — | — | — | — | | |

3 hr 26 min instead of 4 hrs

+ Supervisory overhead !

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Console debugging:

1. elim. of queuing of programs for machine ← main reason
2. Continuity of programmer effort
3. Prompt recog. of useless activity
4. Prompt exploitation of produced results
5. More selective post-mortem info.

debugging passed by user not production work
10 runs per day instead of 5 days etc

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On-line Demands:

using uniprogramming - every problem would have to be finished in response time of highest priority.

Man-Machine Cooperation.

Human judgment – very hard to program, or requiring a large no. of cases.
eg. analog computers.

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Multiprog Standards:

1. each prog. indep. written & compiled
2. minimum of machine's features are placed beyond the programmers direct influence
3. Amt. of additional info. required from programmer – held to a minimum
4. No program should be allowed to introduce error or undue delay into another program.
5. Machine malfunctions and programming errors which cause internal alarms should be analyzed by supervisor prog. to determine which prog. & communicate to man!
6. Routine scheduling done by supervisor – can be overridden by operating staff.
7. Allocation of storage & I/O by needs of programs being done.
8. accounting for machine time done by supervisor.